

N-Body Initial Values Conditioner

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1 Introduction

In the N-body problem we don't have the solution explicitly in general, so we need to approximate it numerically and therefore we need to choose initial conditions that we want to simulate. In many cases we may think in astrophysics terms, using density profiles like Plummer, Hernquist *etc.* But sometimes we may need just specific things from the problem, like having some values for the first integrals or start in (virial) equilibrium.

This package do this: given state vectors $\mathbf{m} \in \mathbb{R}^N$, $\mathbf{q} \in \mathbb{R}^{3N}$ and $\mathbf{p} \in \mathbb{R}^{3N}$, it applies a conditioner that moves the vectors to state vectors that correspond to what we want, like having the total energy $E = 0$, as long as it's physically possible.

For now, we can condition the state vectors to have the desired first integrals and to start in equilibrium.

2 Some concepts

In the N-body problem (in 3d) we have 10 first integrals, being:

- The total energy E ;
- The total angular momentum $\mathbf{J} \in \mathbb{R}^3$;
- The total linear momentum $\mathbf{P} \in \mathbb{R}^3$;
- The “center of mass” $\mathbf{G} \in \mathbb{R}^3$.

The quotation marks is due to $\mathbf{G}$ not being the actual center of mass, but a kind of reference frame:

$$\mathbf{G} = M\mathbf{q}_{com} - t\mathbf{P}, \quad M = \sum_{a=1}^N m_a. \quad (1)$$

A good thing about the N-body problem is that it's potential is invariant under translations, so we ever start with $\mathbf{q}_{com} = \mathbf{0}$ without any loss of generality. The translation that we need to do with some state vectors to obtain $\mathbf{q}_{com} = \mathbf{0}$ is

$$\mathbf{q}_a^{\text{new}} = \mathbf{q}_a - \mathbf{q}_{com}, \quad a = 1, 2, \dots, N. \quad (2)$$

It's evident that any homothety over the new positions doesn't affect the new c.o.m., so this is made only one time at the begin of every method.

Another observation we need to do is that the classic potential energy

$$V = - \sum_{b < a} G \frac{m_a m_b}{\|\mathbf{q}_b - \mathbf{q}_a\|} = - \sum_{b < a} G \frac{m_a m_b}{r_{ab}} \quad (3)$$

is **homogeneous** with degree -1 . However, when we consider the softened problem, the softened potential energy isn't homogeneous anymore:

$$V_\varepsilon = - \sum_{b < a} G \frac{m_a m_b}{\sqrt{r_{ab}^2 + \varepsilon^2}}. \quad (4)$$

3 An iterative method to condition the first integrals

A first attempt may be to condition all the first integrals separately.

3.1 Total linear momentum

Starting by the total linear momentum $\mathbf{P}$, we can apply the same idea of the c.o.m., but we need to choose a weight vector $\mathbf{w} \in \mathbb{R}^N$ and find some vector $\mathbf{W}_P$ such that

$$\sum_{a=1}^N \mathbf{p}_a + w_a \mathbf{W}_P = \mathbf{P} + \mathbf{W}_P = \tilde{\mathbf{P}} \Rightarrow \mathbf{W}_P = \tilde{\mathbf{P}} - \mathbf{P}, \quad (5)$$

where $\tilde{\mathbf{P}}$ is the desired total linear momentum and we have

$$\mathbf{p}_a^{\text{new}} = \mathbf{p}_a + w_a(\tilde{\mathbf{P}} - \mathbf{P}). \quad (6)$$

The simplest choose may be $w_a = m_a/M$.

3.2 Total angular momentum

For the total angular momentum $\mathbf{J}$, the thing is a little more complicated. Although we developed it independetly, Aarseth already used the idea in a two-body problem [1, p. 135].

Consider a particle with position $\mathbf{r}$ and a vector $\boldsymbol{\omega}$ that defines a rotation axis from which distante to $\mathbf{r}$ is ρ and forms an angle ξ . Consider that the velocity of the particle wrt the axis $\boldsymbol{\omega}$ is $\mathbf{v}$ (see figure 1).

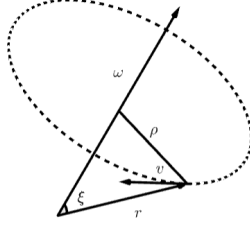


Figure 1: Representation of a particle and a rotation axis.

In this situation, we have that $\|\mathbf{v}\| = \rho \|\boldsymbol{\omega}\|$ and $\sin \xi = \rho / \|\mathbf{r}\|$. Because $\mathbf{v} \perp \boldsymbol{\omega}$ and $\mathbf{v} \perp \mathbf{r}$, we can conclude that $\mathbf{v} \parallel (\mathbf{r} \times \boldsymbol{\omega})$. More than this,

$$\|\mathbf{r} \times \boldsymbol{\omega}\| = \|\mathbf{r}\| \|\boldsymbol{\omega}\| \sin \xi = \|\mathbf{v}\| \quad \therefore \mathbf{v} = \mathbf{r} \times \boldsymbol{\omega}. \quad (7)$$

The angular momentum $\mathbf{J}$ is perpendicular to $\mathbf{r}$ and $\mathbf{v}$ too, but $\mathbf{J}$ and $\boldsymbol{\omega}$ are collinear only when $\mathbf{r} \perp \boldsymbol{\omega}$. However, we can construct an operator $\mathbf{W} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that takes $\boldsymbol{\omega}$ to $\mathbf{J}$:

$$\mathbf{W} = m \begin{bmatrix} -(r_2^2 + r_3^2) & r_1 r_2 & r_1 r_3 \\ r_1 r_2 & -(r_1^2 + r_3^2) & r_2 r_3 \\ r_1 r_3 & r_2 r_3 & -(r_1^2 + r_2^2) \end{bmatrix}. \quad (8)$$

This operator is called the **inertia tensor** (although is not really a *tensor*), and in the literature it appears with the opposite sign sometimes. We can write the angular momentum with:

$$\mathbf{J} = \mathbf{W}\boldsymbol{\omega}. \quad (9)$$

With more than one particle, we can define a common rotation axis $\boldsymbol{\omega}$ and from it apply transformations over the angular velocity of each body. Let $\mathbf{W}_T = \sum_{a=1}^N \mathbf{W}_a$. To obtain the desired $\tilde{\mathbf{J}}$, we need to find the $\boldsymbol{\omega}$ such that

$$\mathbf{W}_T \boldsymbol{\omega} = \tilde{\mathbf{J}} - \mathbf{J} \quad (10)$$

and apply the transformation

$$\mathbf{p}_a^{\text{new}} = \mathbf{p}_a + m_a \mathbf{q}_a \times \boldsymbol{\omega}. \quad (11)$$

3.3 Classic total energy

In the classic total energy we have the homogeneous potential, so we can just look for α and β and define $\mathbf{q}^{\text{new}} = \alpha^{-1} \mathbf{q}$ and $\mathbf{p}^{\text{new}} = \beta \mathbf{p}_a$ such that

$$T(\beta \mathbf{p}) + V(\alpha^{-1} \mathbf{q}) = \beta^2 T + \alpha V = \tilde{E}. \quad (12)$$

We have many forms to solve this equation. If $\tilde{E} \geq 0$, we can just set $\alpha = 1$ and obtain

$$\beta = \pm \sqrt{\frac{\tilde{E} - V}{T}}. \quad (13)$$

In other cases, we can just choose

$$\alpha = 1 + \frac{\tilde{E}}{V}, \quad \beta = \sqrt{-\frac{V}{T}}. \quad (14)$$

3.4 Softened total energy

In the softened case, if the desired total energy is greater than zero the process is the same, as long we don't need to change the positions. If $E < 0$, we can apply the same factor over the velocities:

$$\beta = \sqrt{-\frac{V}{T}}, \quad (15)$$

and it gives a kinetic energy $\tilde{T} = \beta^2 T(\mathbf{p})$. What we need to obtain $\tilde{E}$ is a factor α such that $V_\varepsilon(\alpha^{-1} \mathbf{q}) + \tilde{T} = \tilde{E}$. Being $\alpha = 1/\varepsilon\mu$, we have

$$\varepsilon^{-1} V_1(\mu \mathbf{q}) + \tilde{T} - \tilde{E} = 0. \quad (16)$$

Let $f(\mu)$ the left side. The problem is to find a root μ for this equation, and it can be done by the Newton-Raphson method, given

$$f'(\mu) = \varepsilon^{-1} \sum_{b < a}^N \frac{G m_a m_b r_{ab}^2}{(r_{ab}^2 \mu^2 + 1)^{3/2}}. \quad (17)$$

3.5 The algorithm

It's evident that the applications impact each other. What we can do is define the application $A_{\text{iter}} = \iota_E \circ \iota_J \circ \iota_P$ and suppose that this have a fixed point, so we can apply it some times and get a small error in the final result.

The algorithm is then:

1. Move the c.o.m. to the origin;
2. Apply the total linear momentum conditioner (6).
3. Apply the total angular momentum conditioner (11).
4. Apply one of the total energy conditioners.
5. If the error on the first integrals isn't acceptable, go back to 2.

4 A direct method to condition the first integrals without softening

In the homogeneous case (without softening), we can incorporate all the conditioners into one direct conditioner, i.e., it doesn't need iterations and have machine precision. Observe that, as everywhere, we consider the c.o.m. at origin.

To conditionate the momenta constants we can just unify the two methods, considering the weights the normalized masses:

$$\mathbf{p}_a^{\text{new}} = \mathbf{p}_a - \frac{m_a}{M}(\mathbf{P} - \tilde{\mathbf{P}}) - m_a \mathbf{q}_a \times \boldsymbol{\omega}, \quad \mathbf{W}_T \boldsymbol{\omega} = \mathbf{J} - \tilde{\mathbf{J}}. \quad (18)$$

For the energy, we also use homothety, but because applying $\mathbf{p} \mapsto \beta \mathbf{p}$ affects the momenta constants and $\mathbf{q} \mapsto \alpha^{-1} \mathbf{q}$ affects the angular momentum, we need to include these constants:

$$\mathbf{p}_a = \beta \left(\mathbf{p}_a - \frac{m_a}{M} \left(\mathbf{P} - \frac{1}{\beta} \tilde{\mathbf{P}} \right) - m_a \mathbf{q}_a \times \boldsymbol{\omega} \right), \quad \mathbf{W}_T \boldsymbol{\omega} = \mathbf{J} - \frac{\alpha}{\beta} \tilde{\mathbf{J}}. \quad (19)$$

Fixing α , we have

$$\beta = \pm \sqrt{-\frac{V + S_2}{S_1}}, \quad (20)$$

where

$$S_1 = T - \frac{1}{2M} \|\mathbf{P}\|^2 + \frac{1}{2} \langle \boldsymbol{\omega}_0, \mathbf{J}^0 \rangle, \quad \mathbf{W}_T \boldsymbol{\omega}_0 = \mathbf{J}, \quad (21)$$

$$S_2 = \frac{1}{2} \|\tilde{\mathbf{P}}\|^2 - \frac{\alpha^2}{2} \langle \tilde{\boldsymbol{\omega}}, \tilde{\mathbf{J}} \rangle, \quad \mathbf{W}_T \tilde{\boldsymbol{\omega}} = \tilde{\mathbf{J}}. \quad (22)$$

The process to obtain this simplification is in the Appendix A.

5 Starting in equilibrium

To standardize the numerical study of the N-body problem, Heggie proposed some default values [2]:

$$E = -1/4, \quad R_V = 1, \quad G = 1, \quad M = 1, \quad (23)$$

where $R_V \approx -GM^2/(2V)$ is called the **virial radius**. The word *virial* is related to the equilibrium that occurs when the second derivative of the moment of inertia

$$I = \sum_{a=1}^N m_a \|\mathbf{q}_a\|^2 = -\frac{1}{2} \text{Tr } \mathbf{W}_T, \quad (24)$$

that is given by the Lagrange-Jacobi identity:

$$\ddot{I} = 4E - 2 \sum_{a=1}^N \left\langle \mathbf{q}_a, \frac{\partial V}{\partial \mathbf{q}_a} \right\rangle, \quad (25)$$

that can be simplified in the case of homogeneous potential by the Euler's theorem for homogeneous functions:

$$\ddot{I} = 4E - 2V. \quad (26)$$

To construct initial conditions with these default values, we apply the same process to condition the first integrals, normalize the masses and apply a one more conditioner: letting $Q_v = \sqrt{-V/2T}$, we take $\beta = V/(2E) = -2V$ and apply:

$$\mathbf{q}^{\text{new}} = \beta \mathbf{q}, \quad \mathbf{p}^{\text{new}} = \frac{Q_V}{\sqrt{\beta}} \mathbf{p}. \quad (27)$$

5.1 Softened potential for negative energies

With the loss of homogeneity, we need to use the equation (25), and we can do something related to what we did for the iterative method: we apply (using V_ε)

$$\mathbf{p}^{\text{new}} = \frac{Q_V}{\sqrt{\beta}} \mathbf{p}. \quad (28)$$

To conditionate the virial radius we need to have

$$T(\mathbf{p}^{\text{new}}) = \tilde{E} - V_\varepsilon(\alpha^{-1} \mathbf{q}). \quad (29)$$

We can define the following function f :

$$\begin{aligned} f(\alpha) &= 2\tilde{E} - V_\varepsilon(\alpha^{-1} \mathbf{q}) + \sum_{a=1}^N \left\langle \mathbf{F}_a^{(\varepsilon)}(\alpha^{-1} \mathbf{q}), \alpha^{-1} \mathbf{q}_a \right\rangle \\ &= 2\tilde{E} - 2\alpha V_{\varepsilon\alpha}(\mathbf{q}) + \alpha^2 \sum_{a=1}^N \left\langle \mathbf{F}_a^{(\alpha\varepsilon)}(\mathbf{q}), \mathbf{q}_a \right\rangle. \end{aligned} \quad (30)$$

Again, we need to use an iterative method to evaluate α , like the Newton-Raphson method. For this, being $\mu = \alpha^{-1}$ we get:

$$f'(\mu^{-1}) = 2\mu^{-2} V_{\varepsilon/\mu}(\mathbf{q}) - \frac{2}{\mu} \frac{dV_{\varepsilon/\mu}}{d\mu} - \frac{1}{\mu^2} \sum_{a=1}^N \left\langle \mathbf{F}_a^{(\varepsilon/\mu)}(\mathbf{q}), \mathbf{q}_a \right\rangle + \frac{1}{\mu} \sum_{a=1}^N \left\langle \frac{d\mathbf{F}_a^{(\varepsilon/\mu)}(\mathbf{q})}{d\mu}, \mathbf{q}_a \right\rangle. \quad (31)$$

The derivative of forces wrt μ are

$$\frac{d\mathbf{F}_a^{(\varepsilon/\mu)}(\mathbf{q})}{d\mu} = \frac{3\varepsilon^2}{\mu^3} \sum_{b \neq a} G m_a m_b \frac{\mathbf{q}_b - \mathbf{q}_a}{(r_{ab}^2 + (\varepsilon/\mu)^2)^{5/2}}, \quad (32)$$

and of the potential is

$$\frac{dV_{\varepsilon/\mu}}{d\mu} = -\frac{\varepsilon^2 G}{\mu^3} \sum_{a < b} \frac{m_a m_b}{(r_{ab}^2 + (\varepsilon/\mu)^2)^{3/2}}. \quad (33)$$

As the initial candidate we can take $\mu_0 = -2V_\varepsilon(\mathbf{q})$, that is the solution for the classical problem. At the end, we have a potential that with the desired $\tilde{E}$ guarantee equilibrium. The kinetic energy is not the desired, so we re-conditionate the velocities:

$$\tilde{\mathbf{p}} = \left(\frac{V(\alpha^{-1}\mathbf{q})}{\tilde{E}} - 1 \right)^{1/2} \mathbf{p}. \quad (34)$$

A Simplifying the direct method

Fixing α , we get

$$\beta = \sqrt{-\frac{V}{T(\mathbf{p}^{\text{new}}/\beta)}}. \quad (35)$$

Expanding:

$$\begin{aligned} T(\mathbf{p}^{\text{new}}/\beta) &= \sum_{a=1}^N \frac{1}{2m_a} \left\| \mathbf{p}_a - \frac{m_a}{M} \left(\mathbf{P} - \frac{1}{\beta} \tilde{\mathbf{P}} \right) - m_a \mathbf{q}_a \times \boldsymbol{\omega} \right\|^2 \\ &= \sum_{a=1}^N \frac{1}{2m_a} \left\| \left(\mathbf{p}_a - \frac{m_a}{M} \mathbf{P} - m_a \mathbf{q}_a \times \mathbf{W}_T^{-1} \mathbf{J} \right) + \frac{1}{\beta} \left(\frac{m_a}{M} \tilde{\mathbf{P}} + \alpha m_a \mathbf{q}_a \times \mathbf{W}_T^{-1} \tilde{\mathbf{J}} \right) \right\|^2 \\ &= \sum_{a=1}^N \frac{1}{2m_a} \left(\|\mathbf{K}_a^1\|^2 + \frac{1}{\beta^2} \|\mathbf{K}_a^2\|^2 + \frac{2}{\beta} \langle \mathbf{K}_a^1, \mathbf{K}_a^2 \rangle \right). \end{aligned} \quad (36)$$

Although it looks more complicated, we can simplify it a lot. First,

$$\sum_{a=1}^N \frac{1}{2m_a} \langle \mathbf{K}_a^1, \mathbf{K}_a^2 \rangle = 0. \quad (37)$$

Now, being $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, it's vector product can be write in a matrix form:

$$\mathbf{u} \times \mathbf{v} = [\mathbf{u}]_{\times} \mathbf{v}, \quad [\mathbf{u}]_{\times} = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}. \quad (38)$$

This matrix has the following property:

$$[\mathbf{u}]_{\times}^T [\mathbf{u}]_{\times} = \|\mathbf{u}\|^2 \mathbf{Id}_3 - \mathbf{u}\mathbf{u}^T, \quad (39)$$

so we can write

$$\|\mathbf{u} \times \mathbf{v}\|^2 = \mathbf{v}^T (\|\mathbf{u}\|^2 \mathbf{Id}_3 - \mathbf{u}\mathbf{u}^T) \mathbf{v}. \quad (40)$$

Therefore we can write the inertia tensor for a body as

$$\mathbf{W}_a = m_a (\mathbf{q}_a \mathbf{q}_a^T - \|\mathbf{q}_a\|^2 \mathbf{Id}_3), \quad (41)$$

and

$$\sum_{a=1}^N \|\mathbf{q}_a \times \mathbf{W}_T^{-1} \mathbf{J}\|^2 = -\mathbf{J}^T \mathbf{W}_T^{-T} \mathbf{W}_T \mathbf{W}_T^{-1} \mathbf{J} = -\mathbf{J}^T \mathbf{W}_T^{-1} \mathbf{J}. \quad (42)$$

The simplification is trivial from here.

References

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